

Polynomial order for permutations containing exactly one 1432

Report 139

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Abstract

Let b_n count permutations with exactly one classical occurrence of 1432. Reversal identifies this sequence with OEIS A224182, which is defined using 2341. We prove $b_n = \Theta(9^n/n^3)$ by two explicit injections and a minima-preserving Ferrers-board correspondence. The upper injection has a computer-free eight-case proof and an exact marked inverse. The lower injection inflates a left-to-right minimum; an entropy estimate shows that such minima have asymptotic mean $n/3$ in the avoidance class. If A_n is the number of 1234-avoiders, the bounds give normalized lower and upper constants $1/2187$ and 81 . They imply an inverse-threshold bracket of unrounded asymptotic width $11/2$. An additional, explicitly computer-assisted result characterizes the eligible marked avoiders by adjacency and four singleton rectangles. No leading equivalent, amplitude limit, or proposed ratio $3/80$ is proved.

[write] **Status and trust boundary, added 5 October 2026, batch 105.** This is bundle Report 139 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a224182-unique-1432-order** (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.1). It is unrefereed and not formalized; its author line reads “Report 139” and its PDF author field is empty. *What rests on what.* Theorem 1.1 (the order $b_n = \Theta(9^n/n^3)$ with the constants $1/2187$ and 81) has a computer-free proof; it cites Regev’s strip asymptotic [7] and Krattenthaler’s growth-diagram theorems [6], which are not re-proved. Theorem 7.1 (the exact image of the split) is computer-assisted: a proved reduction to at most eight points, then an exhaustive finite certificate of 2,074 designated cores, replayed by two independent programs (Section 7). No asymptotic statement rests on a computation. The write adds three remarks with proofs: Remark 6.2 (the inverse as an instance of the transseries volume’s Lambert core), Remark 6.3 (the generating function of A224182 is not algebraic) and Remark 8.1 (the one-step bound that the source attributes to Mansour, Rastegar and Roitershtein is not justified by their printed argument, which gives a factor n^2 instead of n , one power of n short of the upper order), and Section 9.1, the open questions with a recomputation of $b_n/(nA_n)$ for $n \leq 18$.

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1 The result and its scope

An occurrence of a classical pattern is a subsequence, with no adjacency requirement, whose relative value order is the pattern. Put

$$b_n = \#\{\pi \in S_n : N_{1432}(\pi) = 1\}, \quad A_n = |\text{Av}_n(1432)|, \quad M_n = \sum_{\pi \in \text{Av}_n(1432)} \text{lmin}(\pi),$$

where $\text{lmin}(\pi)$ is the number of left-to-right minima. Set $A_0 = 1$, $M_0 = 0$, and $b_n = 0$ for $n < 4$. We prove below that $A_n = |\text{Av}_n(1234)|$, including equality of the full minima-skeleton refinements. These are equal cardinalities, not equal avoidance sets. The classical three-row strip asymptotic of Regev [7] is

$$A_n \sim C_0 \frac{9^n}{n^4}, \quad C_0 = \frac{81\sqrt{3}}{16\pi}. \quad (1)$$

Theorem 1.1. *For every $n \geq 4$,*

$$M_{n-3} \leq b_n \leq (n+2)A_{n+2}. \quad (2)$$

Moreover, $M_m \sim (m/3)A_m$. Consequently,

$$b_n = \Theta(9^n n^{-3}), \quad \log b_n = n \log 9 - 3 \log n + O(1), \quad (3)$$

$$\frac{1}{2187} \leq \liminf_{n \rightarrow \infty} \frac{b_n}{nA_n} \leq \limsup_{n \rightarrow \infty} \frac{b_n}{nA_n} \leq 81. \quad (4)$$

All assertions of this theorem have a computer-free proof.

Reversal sends 2341 to 1432, preserving the number of occurrences; thus b_n is A224182 [1]. Nakamura [2, Section 3.2, Theorem 4] gave a catalytic functional equation for 2341 and exact initial coefficients. Conway and Guttmann [3, Section 5.3] conjectured the stronger equivalent $b_n \sim C_1 9^n / n^3$, with $C_1/C_0 \approx 0.0375$ and the candidate $C_1/C_0 = 3/80$. Theorem 1.1 proves the polynomial order in that prediction, but not convergence of $b_n/(nA_n)$. The proposed $3/80$ is a ratio of amplitudes; it is not the absolute prefactor C_1 .

The shape-Wilf ingredient is due to Backelin, West and Xin [5]; the directly inspected growth-diagram proof used here is Krattenthaler [6, Theorems 1–3]. The split is adapted from Burstein’s construction for a single global 321 [4]; a supported decreasing triple requires the separate proof in Section 5. Known fixed-occurrence exponential-growth results are discussed in Section 8. No worldwide priority claim is made.

1.1 Provenance, sources and reading conventions

[write] **Added 5 October 2026, batch 105.** *Provenance.* This report is Report 139 of the session bundle of Reports 1–243 that ProveIt’s docs/incoming intake processed as batch 105: the archive A224182_Polynomial_Growth_Order_and_Inverse_Bounds_Source.zip (1,316,107 bytes, wrapper directory Report139/, 10 files), arrival commit 60f54ea06, placed in e85586b7c. The text is the delivered Report139.tex (707 lines), shipped as article.tex. The delivered 10-page PDF and the checksum manifest manifest.json are not shipped (the manifest was verified at placement); both are retrievable from the arrival commit. The package names no ProveIt commit, path or report, so the report has no pin; it carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 34 delivered labels with u1432: and updated every reference; it added this subsection, the notes marked [write], Remarks 6.2, 6.3 and 8.1, Section 9.1, a table of contents and two bibliography entries, and typeset three binomial coefficients with \binom instead of \choose (identical output; this removes an amsmath warning). No statement, proof or number of the source was changed: the added remarks are the last statements of their sections and the added displays are unnumbered, so every theorem and equation keeps its delivered number.

The sources, as the write read them (5 October 2026).

- OEIS A224182 [1] (revision #11 of September 3, 2026): “Number of permutations of length n containing exactly 1 occurrence of 2341”, offset 1, 18 terms, through $b_{18} = 2,055,119,789,289$; Nakamura is its author. Reversal turns 2341 into 1432.
- Nakamura [2]: Section 3.2 extends the 231 method to 2341, and Theorem 4 there is the functional equation (FE2341); located.
- Conway and Guttmann [3]: Section 5.3 analyses the 18 OEIS terms of A224182 together with 26 further approximate terms of their own, finds the ratio $\psi_1^{III}(n)/(n\psi_0^{III}(n))$ (here $b_n/(nA_n)$) “going to an apparent limit of 0.0375 ± 0.0005 ”, and writes that it is tempting to guess that this is $3/80$; they conclude $b_n \sim C_1 9^n/n^3$ with $C_1 \approx 0.0375 \cdot 81\sqrt{3}/(16\pi)$, “and perhaps $C_1 = 3^5\sqrt{3}/(2^8 \cdot 5\pi)$ exactly”, which is $(3/80)C_0$. Their Section 5.5 does name A224182 for one occurrence of 1324 (class V), as Section 8 says; so does their Section 4, which attributes those 17 terms to Johansson and Nakamura. Both are slips of the paper: A224182 is the class-III sequence.
- Mansour, Rastegar and Roitershtein [8]: Theorem 3.1 and its proof (arXiv version 1, pp. 28–29; the same text in the published author manuscript, PubMed Central PMC7144682). The display quoted in Section 8 is printed there; Remark 8.1 explains why its printed justification gives a factor n^2 instead of n .
- Bóna and Burstein [9]: Lemma 6.3 (p. 8; its rigorous proof is credited there to Alin Bostan), used in Remark 6.3. Their non-algebraicity theorem (Theorem 6.2) concerns one copy of $k \cdots 21$, k even; their generalized injection (Theorem 4.1) and non-rationality theorem (Theorem 5.5) concern $321 \ominus \rho$, which for length four is only 4321. Neither covers 1432 or any pattern in its symmetry class $\{1432, 2341, 3214, 4123\}$.
- Krattenthaler [6]: Theorems 1–3 are in Section 2 of the arXiv version, as Proposition 2.2 uses them (the growth-diagram bijection, Greene-type chain readings, conjugation swapping the NE and SE chain statistics). The write located them and did not re-derive them. Burstein [4] (three pages) is the short proof of Noonan’s count of permutations with one 321; located.
- Not read by the write: Regev [7] and the journal text of Backelin, West and Xin [5] (which the source also could not retrieve). For Regev’s constant, exact values of A_n from the hook-length formula give $n^4 A_n/(C_0 9^n) = 0.8975, 0.9469, 0.9730, 0.9864$ at $n = 50, 100, 200, 400$, consistent with (1) (an observation, not a check of the cited theorem).

What was checked at intake. The batch-105 dossier read the manuscript in full and found no error. It checked by hand the eight cases of Lemma 5.1, the inflation interval argument and its τ_k version, the entropy maxima $f(1/3) = \log 9$ and $2 \log k$ at $u = 1/k$, the constants $1/2187, 81$ and $11/2$, the monotonicity (15) and the restriction reduction in Theorem 7.1. Its own programs checked the split exhaustively for $n \leq 10$ (two

sizes beyond the companion), the bounds (2) for $n \leq 9$, the brute-force counts b_n for $n \leq 10$ against the OEIS, the collision and sole-support examples and the 106 versus 107 board counts. The delivered suites passed on copies (Section 8, note at the end).

Relation to the repository. Before batch 105 no file of the repository mentioned A224182, A005802 or the papers of Conway–Guttmann, Regev, Bóna–Burstein or Backelin–West–Xin, and the source makes no claim about the repository, so nothing needed correcting. The report meets the following texts.

- **a217057-unique-pattern-occurrences** (same collection, same batch): exactly one 1234 (A217057), one 1243 (A224179) and one 12345 (A224248), from bundle Reports 134–138 and 140. *See also:* its normalizer $A_n = |\text{Av}_n(1234)|$ and Regev’s constant C_0 are the ones here, and Report 136 there derives non-algebraicity of the A224179 generating function from the same Lemma 6.3 of [9] that Remark 6.3 uses. Otherwise the methods share nothing: the reports there work with a tableau spine, three-row strips and unitary integrals and prove amplitudes; this one uses minima skeletons, Ferrers boards and a Burstein-type split and proves an order. Neither source cites the other.
- **a273821-first-pattern-failure** (collection `enumerative-combinatorics`), Part II, Section “A worked example: first failure of 1432” (`fpf:lp:sec:example`): the patterns $\tau_r = 1 \oplus \delta_r$ of Corollary 4.2, studied there through a different statistic (the first failure in 123-avoiders). No shared theorem.
- **a047874-long-increasing-subsequences**: $A_n = |\text{Av}_n(1234)|$ is the number of permutations of $[n]$ whose longest increasing subsequence has length at most three, a partial row sum of A047874. No shared theorem.
- The transseries volume
`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`
(`transseries_and_inversion.tex`): Corollary 6.1 is an instance of its `p0:thm:lambert-core` and fits the setting of its `p0:thm:staircase(1)` (Remark 6.2).

No Lean or Rocq file of the repository treats pattern occurrences; no statement of this report is formalized, and its place in the collection confers no formal status.

[write] What is not claimed, collected from the source. No leading equivalent, no convergence of $b_n/(nA_n)$, no proof of the candidate $3/80$ (which is C_1/C_0 , not C_1), no D-finiteness; the two $o(1)$ errors in the bounds are not asserted equal or effective. No matching upper bound for $1 \oplus \delta_k$, $k \geq 4$. No novelty is claimed for the upper order, and no worldwide priority claim is made; the literature and repository check was “scoped, not an exhaustive worldwide literature review”. Theorem 7.1 is computer-assisted; the main theorem is not. The inverse bracket’s width $11/2$ concerns unrounded endpoints, “not an exact count of admissible integers or a certified finite numerical threshold”. No limiting density of eligible centres. The Backelin–West–Xin full text was not retrieved; the cited Burstein paper does not by itself prove Lemma 5.1. Neither the numerical cross-checks nor the archive seal are formal verification, a digital signature or proof of an amplitude; “no repository publication, outside contact, or external submission” is part of the delivery. The write adds: its numerical observations (Section 9.1) are exact ratios of OEIS terms or floating-point diagnostics, and certify nothing about limits.

Reading conventions (letters the manuscript reuses, with the tempting false readings; no symbol was renamed).

Symbol	Meaning here (where)	Not to be read as
b_n	permutations of $[n]$ with exactly one 1432, OEIS A224182 (offset 1; A224182 itself is stated for 2341)	the 1243 count A224179, which Report 136 (filed in a217057) also writes b_n
A_n	$ \text{Av}_n(1432) $, proved equal to $ \text{Av}_n(1234) $ (A005802)	equality of avoidance <i>sets</i> ; Section 2 proves equal cardinalities only
$N_{1432}(\pi)$, $N(x)$	the number of occurrences in π ; the threshold $\min\{n \geq 4 : b_n \geq x\}$ (Section 6)	each other
M_n , lmin	M_n the total number of left-to-right minima over $\text{Av}_n(1432)$; $\text{lmin}(\pi)$ their number in π	the marks $M(\sigma)$ of Report 136 in a217057
C_0 , C_1 , $C_{\pm}$	Regev's constant $81\sqrt{3}/(16\pi)$; Conway and Guttmann's conjectured amplitude of b_n ; the proved bounds (14)	C_1 as proved; the conjectured $3/80$ is C_1/C_0 , not C_1
a, b, c, d	values of the unique occurrence, $a < d < c < b$, at positions $i < j < k < l$ (Section 5)	
B , C , D , X , Y	the special points of the split (Lemma 5.1); in the marked inverse, $C = q_K$, $D = q_{K-1}$, $B = q_{K+1}$	the constants $C_{\pm}$, C_0 ; $D_{m,r}$
k	a core position (Sections 5, 7); the tail length of τ_k (Corollary 4.2)	each other
K	the marked position in a split image	the tail length k
δ , δ_k	a tolerance in Section 3; δ_k the decreasing permutation of length k	
$D_{m,r}$, H , f	avoiders with r minima; the binary entropy; the entropy exponent (Section 3)	
λ , L , T_C	$\log 9$; $\log x$; the continuous inverse (16)	L as a length
(R1)–(R4)	the four forced singleton regions, equations (10)–(13)	
$\psi_r^{III}(n)$	Conway and Guttmann's name for the number of permutations with r occurrences of a class-III pattern; $b_n = \psi_1^{III}(n)$, $A_n = \psi_0^{III}(n)$	

2 Minima skeletons and Ferrers boards

A left-to-right minimum is an entry smaller than every entry preceding it. Its position and value together are a point of the permutation plot, with position increasing rightward and value upward. Fix an admissible set S of all these points, the *minima skeleton*. Remove their rows and columns. A remaining position h and remaining value y form an allowed cell precisely when

$$(s, t) \in S \text{ exists with } s < h \text{ and } t < y. \quad (5)$$

In the increasing orders of the remaining positions and values, this is a northeast-closed Ferrers board: if a cell is allowed, every remaining cell to its right and above it is allowed.

Lemma 2.1. *The permutations with exact minima skeleton S are in bijection with full matchings of this board, one point in each remaining row and column.*

Proof. Every nonminimum has a smaller earlier minimum and so occupies an allowed cell. Conversely, every point of a matching has an earlier smaller skeleton point, and is therefore not a new minimum. Nor can it invalidate a later skeleton minimum: it is above its earlier supporting minimum, which is

itself above every subsequent skeleton minimum. Restoring S therefore gives exactly the specified skeleton. $\square$

A chain is *board-contained* if its smallest axis-parallel bounding rectangle consists of allowed cells. If three matching points have positions $h_1 < h_2 < h_3$ and values $x > y > z$, the decreasing chain is board-contained exactly when (h_1, z) is allowed. That is exactly the condition that some minimum can support it on the left and below. More precisely, if $\text{supp}_\pi(T)$ counts *all* permutation points before h_1 and below z , then

$$N_{1432}(\pi) = \sum_{T \text{ a contained decreasing triple}} \text{supp}_\pi(T). \quad (6)$$

Indeed every 1432 has a nonminimum decreasing tail. Conversely an arbitrary supporting point has an earlier-or-equal minimum still below z , so its tail is contained. The sum must not count only supporting skeleton points: ordinary points can support the same tail too.

For an increasing triple of remaining points, its first point is both leftmost and lowest. Its cell is allowed, so its entire bounding rectangle is allowed. Such a triple has a smaller earlier minimum and extends to 1234. Conversely the last three entries of any 1234 are nonminima. We have proved

$$\begin{aligned} \pi \in \text{Av}(1432) &\iff \text{the matching has no contained decreasing triple,} \\ \pi \in \text{Av}(1234) &\iff \text{the matching has no increasing triple.} \end{aligned}$$

Proposition 2.2. *For each fixed exact minima skeleton S , the numbers of 1432-avoiders and 1234-avoiders realizing S are equal.*

Proof. Rotate the compressed board through 180 degrees to obtain a southwest Ferrers shape, the French convention of Krattenthaler [6, Section 2]. Rotation acts on a matching by reverse-complement, which fixes both patterns 123 and 321 and preserves rectangle containment.

Here are the precise growth-diagram facts needed from that source. Theorem 1 bijects fillings with at most one point in each row and column with boundary sequences of partitions; occupied rows and columns correspond to strict successive changes. Theorem 2 reads longest increasing and decreasing chains in southwest rectangles from the first row and column of their partition labels. In Theorem 3, conjugating every boundary partition swaps these two maximum-chain statistics. The *decreasing* statistic explicitly requires its bounding rectangle to lie in the shape. Increasing chains in a southwest shape satisfy that requirement automatically, because their northeast endpoint lies in the shape.

Conjugation preserves every strict boundary change, hence preserves occupancy of each row and each column, not merely the total number of points. It therefore restricts to full matchings of our fixed board. Summing the equality of the two chain statistics over the unrestricted maximum and restricting the forbidden maximum to at most two equates avoidance of the two contained triple types. The empty matching is handled separately and trivially. Rotate back and restore the unchanged skeleton. $\square$

Remark 2.3. This is an avoidance correspondence, not an equidistribution of occurrence counts. For the northeast board of six columns with least allowed values $(2, 1, 1, 1, 1, 1)$, the numbers of full matchings with exactly one contained 123 and exactly one contained 321 are respectively 106 and 107. The companion checks these counts directly. Thus a one-occurrence theorem cannot be inferred by transferring one contained chain through shape-Wilf equivalence.

3 An entropy bound for the number of minima

Let $D_{m,r}$ count 1234-avoiders of size m with r left-to-right minima. Delete all their minima. The remainder avoids 123: the first point of any increasing triple in the remainder has a smaller earlier minimum, which would extend it to 1234. Choose the r positions and r values of the minima, whose order is necessarily decreasing, and then the standardized remainder. This is an injective encoding, although some choices do not produce an admissible permutation. Hence

$$D_{m,r} \leq \binom{m}{r}^2 \text{Cat}(m-r) \leq \binom{m}{r}^2 4^{m-r}. \quad (7)$$

Put $H(u) = -u \log u - (1-u) \log(1-u)$, with endpoint convention $0 \log 0 = 0$. The standard binomial entropy bound yields

$$D_{m,r} \leq \exp\{mf(r/m)\}, \quad f(u) = 2H(u) + (1-u) \log 4.$$

On $(0,1)$,

$$f'(u) = 2 \log \frac{1-u}{u} - \log 4, \quad f''(u) = -\frac{2}{u} - \frac{2}{1-u} < 0.$$

Thus f has its unique maximum at $u = 1/3$, where $f(1/3) = \log 9$. For every fixed $\delta > 0$ for which the complement is nonempty, compactness supplies $\eta_\delta > 0$ such that

$$\sum_{|r/m - 1/3| \geq \delta} D_{m,r} \leq (m+1)9^m e^{-\eta_\delta m}.$$

After division by (1), the corresponding probability is $O(m^5 e^{-\eta_\delta m})$. The minima fraction converges to $1/3$ in probability, and its boundedness in $[0,1]$ implies convergence in mean. Proposition 2.2 transfers the entire minima distribution to 1432-avoiders. Therefore

$$M_m \sim \frac{m}{3} A_m. \quad (8)$$

4 A lower injection by inflation

Proposition 4.1. *For $m \geq 1$, there is an injection from 1432-avoiders of size m with a distinguished left-to-right minimum to permutations of size $m+3$ with exactly one 1432.*

Proof. If the marked minimum has value a , replace it by the consecutive block

$$(a, a+3, a+2, a+1),$$

raising every other old value greater than a by three. The four-point block is an interval in both position and value and has pattern 1432.

An occurrence using no block point would have existed before inflation; one using exactly one would contract to an old occurrence. If an occurrence uses two or three block points, these occupy consecutive positions and consecutive ranks *within the selected pattern*. The only proper nonsingleton intervals of 1432 are its tail pairs 43, 32, and its tail triple 432. Thus the first and smallest point of this occurrence would lie outside the block, before it and below it. No such point exists, because the inflated entry was a left-to-right minimum. An occurrence using all four block points is exactly the internal one.

That unique occurrence identifies the inflated interval. Contracting it recovers both the original avoider and its marked minimum, proving injectivity and $M_m \leq b_{m+3}$. $\square$

4.1 A fixed pattern length corollary

The lower construction and minima concentration extend to a descending tail of any fixed length. For $k \geq 2$, put $\tau_k = 1 \oplus \delta_k = (1, k+1, k, \dots, 2)$, and let $A_n^{(k)}$ and $b_n^{(k)}$ count avoidance and exactly one occurrence, respectively.

Corollary 4.2. *For every fixed integer $k \geq 2$,*

$$A_n^{(k)} = |\text{Av}_n(12 \cdots (k+1))|, \quad \liminf_{n \rightarrow \infty} \frac{b_n^{(k)}}{n A_n^{(k)}} \geq k^{-2k-1}.$$

No matching upper bound for general k is asserted.

Proof. The same fixed-skeleton board argument, forbidding contained monotone chains of length k , preserves the minima distribution and proves the avoidance equality. Let r be the number of minima. Deleting them leaves an avoider of $12 \cdots k$. By transposed Robinson–Schensted, each such permutation is a pair of standard Young tableaux with at most $k-1$ rows. A tableau is determined by the row occupied by each of its labels, since entries within a row increase. Encoding both tableaux by these row-label words gives at most $(k-1)^{2(n-r)}$ possibilities, even without imposing the same-shape and standardness conditions on the two words. Consequently

$$D_{n,r}^{(k)} \leq \binom{n}{r}^2 (k-1)^{2(n-r)}.$$

The entropy function $2H(u) + 2(1-u)\log(k-1)$ is strictly concave, uniquely maximized at $u = 1/k$, with value $2\log k$. For this fixed k , Regev’s strip asymptotic has the form $A_n^{(k)} \sim C_k k^{2n} n^{-(k^2-1)/2}$, where $C_k > 0$. The proof in Section 3 therefore gives $M_n^{(k)} \sim (n/k) A_n^{(k)}$.

Inflate a marked minimum into the interval τ_k , increasing the size by k . Every proper nonsingleton interval of τ_k lies wholly in its descending tail: an initial segment containing its first and smallest entry and its largest entry cannot be a proper value interval. Thus an occurrence using between two and k block points would require an outside earlier, smaller supporter, which does not exist. The arguments for zero, one, and all $k+1$ block points are exactly as in Proposition 4.1. The unique internal occurrence recovers the marked source, so $M_{n-k}^{(k)} \leq b_n^{(k)}$. Finally

$$\frac{M_{n-k}^{(k)}}{n A_n^{(k)}} \longrightarrow \frac{1}{k} (k^2)^{-k} = k^{-2k-1}.$$

□

5 The marked split and its eight-case proof

Let the unique occurrence in π have positions $i < j < k < l$ and values $a < d < c < b$. Names of core points will also denote their values when no confusion can result. Write

$$\pi = U b V c W d Z,$$

where the supporting point a lies in U . For $0 < \varepsilon < 1$, form and standardize

$$q = \text{std}(U(c-\varepsilon) V d c b W(c+\varepsilon) Z), \tag{9}$$

marking the retained central c . Thus the *old* d, b move to the immediate positional neighbors of c ; the *new* $c-\varepsilon, c+\varepsilon$ occupy the old b, d slots. For exact integer arithmetic, multiply old coordinates by three and replace the offsets by ± 1 .

5.1 The forced empty regions

Uniqueness of the core forces the following four singleton conditions:

$$\{\pi_t : t < j, \pi_t < d\} = \{a\}, \quad (R1) \quad (10)$$

$$\{\pi_t : i < t < k, \pi_t > c\} = \{b\}, \quad (R2) \quad (11)$$

$$\{\pi_t : j < t < l, d < \pi_t < b\} = \{c\}, \quad (R3) \quad (12)$$

$$\{\pi_t : t > k, a < \pi_t < c\} = \{d\}. \quad (R4) \quad (13)$$

An extra point in each respective region could replace a, b, c, d in the core to give a second occurrence.

Lemma 5.1. *The split permutation q avoids 1432.*

Proof. It is convenient to work in unstandardized plot coordinates. The special points are

$$X = (j, c - \varepsilon), \quad D = (k - \varepsilon, d), \quad C = (k, c), \quad B = (k + \varepsilon, b), \quad Y = (l, c + \varepsilon).$$

Every other point is *ordinary* and unchanged. In their position order the five special points have pattern 21354. The only inverted special pairs are (X, D) and (B, Y) , and C makes no inversion with another special point. Consequently a decreasing triple has at most two special points.

Suppose a 1432 in the output has points (u, v, w, z) in position order, with values $u < z < w < v$. Its *tail* is (v, w, z) . If the tail contains none of X, D, B, Y , it is unchanged from the original. If u is ordinary, the whole occurrence already existed and differs from the core (the old b, d are absent from the unchanged set). If u is special, replace it by a , which is earlier and smaller than every special point. The same tail again gives a distinct original occurrence. Both are impossible.

For the remaining cases use transposition of the plot. The pattern 1432 is its own inverse, and the construction commutes with transposition: X and D are interchanged, as are B and Y , while the first and last tail roles are interchanged. Thus for a tail with one special point it suffices to consider D and B in each of the three roles. All other tail points in these six cases are ordinary.

Case 1. Tail (D, w, z) . Here $u < z < w < d$, so u is ordinary. Since D is immediately before C and w, z are ordinary points after D , they are after old position k . Replacing D by old c gives the distinct original occurrence (u, c, w, z) .

Case 2. Tail (v, D, z) . Again u is ordinary. If $v > c$, replacement of D by old c gives (u, v, c, z) . Otherwise $d < v < c$. If v is after j , it is between j and k , violating (R3). If it is before j , then u is before j and below d , so (R1) forces $u = a$. Now z is after k with $a < z < d < c$, violating (R4).

Case 3. Tail (v, w, D) . The supporter u is ordinary. Replace D by old d at position l ; this gives an original occurrence (u, v, w, d) . It cannot be the core: w is before k , whereas core c is at k .

Case 4. Tail (B, w, z) . If $z > a$, then (a, b, w, z) using old b is a distinct original occurrence. Thus $z < a$, and $u < z < a < d$ implies that u is ordinary before k . If $w < c$, then (u, c, w, z) is an original occurrence. Otherwise $c < w < b$; since w is after k , (R3) forces it to be after l . The point z is then also after l , and (u, c, d, z) is an original occurrence, since $u < z < a < d < c$.

Case 5. Tail (v, B, z) . The ordinary point v is before k and above $b > c$. By (R2) it must precede i , and hence j . Replacing B by old b gives the distinct original occurrence (u, v, b, z) .

Case 6. Tail (v, w, B) . The ordinary point w is before k and above $b > c$. By (R2) it must precede i , and hence j . Replacing B by old b gives the distinct original occurrence (u, v, w, b) .

It remains to consider two special tail points. They must be either (X, D) or (B, Y) . If they occupy the first and last tail roles, the intermediate ordinary point violates (R3): it is between j, k

with value between d, c in the first case, or between k, l with value between c, b in the second. Thus the special pair occupies the first two or last two roles. Transposition exchanges these placements within each pair, so only the first-two placement needs checking.

Case 7. Tail (X, D, z) . The supporter u is ordinary, before j , with $u < z < d$. By (R1), $u = a$. But z is ordinary after k and $a < z < d < c$, contrary to (R4).

Case 8. Tail (B, Y, z) . The ordinary point z is after l and below c , because Y is immediately above c among old values. If $z > a$, it violates (R4). Hence $z < a$, and $u < z < a < d$ shows that u is ordinary before k . Then (u, c, d, z) is a distinct original 1432.

Every case contradicts uniqueness or a forced region condition. $\square$

5.2 The exact marked inverse

Let the marked position in the standardized output be K and its value $C = q_K$. Let $D = q_{K-1}$ and $B = q_{K+1}$. Its adjacent values $C - 1$ and $C + 1$ occur at positions $J < K - 1$ and $L > K + 1$, respectively, and $D < C < B$. Delete the two positional neighbors of K , replace the value at J by B and that at L by D , and standardize. This recovers π exactly. Thus the marked split is injective into the $(n + 2)$ -pointed avoiders of size $n + 2$, proving the upper bound in (2).

The mark is essential. The two distinct sources

$$(3, 5, 4, 6, 8, 1, 9, 7, 2), \quad (3, 8, 5, 1, 4, 2, 6, 9, 7)$$

both split to $(3, 5, 4, 6, 10, 1, 7, 2, 8, 11, 9)$, with respective *one-based* marked positions 9 and 4. Each source has exactly one occurrence. This explicit collision rules out forgetting the mark in this construction.

6 The constants and inverse scale

Combining (8) with Regev's equivalent gives

$$\frac{M_{n-3}}{nA_n} \longrightarrow \frac{1}{3 \cdot 9^3} = \frac{1}{2187}, \quad \frac{(n+2)A_{n+2}}{nA_n} \longrightarrow 9^2 = 81.$$

This proves Theorem 1.1. Equivalently, with

$$C_- = \frac{C_0}{2187}, \quad C_+ = 81C_0, \tag{14}$$

one has

$$(C_- + o(1)) \frac{9^n}{n^3} \leq b_n \leq (C_+ + o(1)) \frac{9^n}{n^3}.$$

The two $o(1)$ errors are not asserted to be equal or effective.

Prepending or appending a new maximum leaves the number of 1432 occurrences unchanged: a new maximum cannot participate in that pattern at the first or last position of the full permutation. Both maps are injective and their images are disjoint. Therefore

$$b_{n+1} \geq 2b_n \quad (n \geq 4). \tag{15}$$

For real $x > 0$, define the finite threshold $N(x) = \min\{n \geq 4 : b_n \geq x\}$. All logarithms below are natural. Put $\lambda = \log 9$, $L = \log x$, and

$$T_C(x) = \frac{L + 3 \log L - 3 \log \lambda - \log C}{\lambda}. \tag{16}$$

Corollary 6.1. *As $x \rightarrow \infty$,*

$$\lceil T_{C_+}(x) + o(1) \rceil \leq N(x) \leq \lceil T_{C_-}(x) + o(1) \rceil. \quad (17)$$

In particular $N(x) = (\log x + 3 \log \log x) / \log 9 + O(1)$. The unrounded endpoints differ by $11/2 + o(1)$.

Proof. For fixed $C > 0$, the eventually increasing continuous function $Ce^{\lambda t}/t^3$ has inverse $T_C(x) + O(\log L/L)$. This follows by substitution in $\lambda t - 3 \log t + \log C = L$; the residual is $O(\log L/L)$ and the derivative tends to $\lambda > 0$.

The two-sided order estimate first gives $N(x) = (L + 3 \log L) / \lambda + O(1)$. Uniformly on any fixed enlargement of this bounded-width index window, the relative errors in (14) tend to zero. Bound their absolute values by one positive number $\epsilon(x) \rightarrow 0$, chosen also to cover the two continuous inverse endpoints. Let t_+ and t_- be the real crossings of x by the upper and lower envelopes $(C_+ + \epsilon(x))e^{\lambda t}/t^3$ and $(C_- - \epsilon(x))e^{\lambda t}/t^3$, respectively. At the integer $\lceil t_+ \rceil - 1$, the upper envelope is strictly below x , so $b_{\lceil t_+ \rceil - 1} < x$. At $\lceil t_- \rceil$, the lower envelope is at least x , so $b_{\lceil t_- \rceil} \geq x$. These two integers lie in the chosen window for large x . Monotonicity gives $\lceil t_+ \rceil \leq N(x) \leq \lceil t_- \rceil$. The inverse formula just established and $\log(C_{\pm} \pm \epsilon(x)) = \log C_{\pm} + o(1)$ give (17).

Finally

$$T_{C_-}(x) - T_{C_+}(x) = \frac{\log(C_+/C_-)}{\log 9} = \frac{\log(81 \cdot 2187)}{\log 9} = \frac{11}{2}.$$

□

The endpoint errors in (17) occur *inside* the ceilings and may differ. They cannot be discarded near an integer. The width $11/2$ describes real endpoints, not an exact count of admissible integers or a certified finite numerical threshold.

Remark 6.2 ([write] The inverse in the terms of the transseries volume, 5 October 2026). The equation $\lambda t - 3 \log t + \log C = L$ inverted in the proof of Corollary 6.1 is the dominant block $\Phi_0(X) = aX + b \log X = L'$ of `p0:thm:lambert-core` in the transseries volume [10], with

$$a = \lambda = \log 9, \quad b = -3, \quad L' = L - \log C.$$

Since $b < 0$, part (3) of that theorem says that for $L' > L_c = 3(1 - \log(3/\lambda))$ there are exactly two positive solutions, and the large one, on which Φ_0 increases, is

$$X = \frac{3}{\lambda} \left(-W_{-1} \left(-\frac{\lambda}{3} e^{-L'/3} \right) \right).$$

With $u = \frac{\lambda}{3} e^{-L'/3} \rightarrow 0^+$, the expansion $-W_{-1}(-u) = \ell_1 + \ell_2 + O(\ell_2/\ell_1)$, $\ell_1 = \log(1/u) = L'/3 + \log(3/\lambda)$, $\ell_2 = \log \ell_1 = \log L - \log 3 + O(1/L)$, gives

$$X = \frac{L' + 3 \log(3/\lambda) + 3 \log L - 3 \log 3}{\lambda} + O\left(\frac{\log L}{L}\right) = T_C(x) + O\left(\frac{\log L}{L}\right),$$

the formula used in the proof, so $T_C(x)$ is the two-term expansion of the Lambert solution. The threshold $N(x)$ is the integer staircase N_* of `p0:def:three-inverses` with $n_1 = 4$: by (15) and $b_4 = 1$, $(b_n)_{n \geq 4}$ is strictly increasing and unbounded, so it has admissible interpolations (piecewise linear, for example), and `p0:thm:staircase` (1) gives $N(x) = \lceil \nu(x) \rceil$ for each of them. The proof of Corollary 6.1 does not choose an interpolation: it compares b_n with two continuous envelopes at the integers, which yields the two-ceiling bracket directly. No novelty is claimed for the inversion.

Remark 6.3 ([write] The generating function is not algebraic, 5 October 2026). The generating function $B(z) = \sum_{n \geq 1} b_n z^n$ of A224182 is not an algebraic power series.

Proof. Put $f(z) = z + z^2 + z^3 + B(z)$, so that $f_n = 1$ for $n \leq 3$ and $f_n = b_n$ for $n \geq 4$. For $n \geq 4$, (2) gives $b_n \geq M_{n-3} \geq A_{n-3} \geq 1$, because every permutation has at least one left-to-right minimum; hence $f_n > 0$ for every $n \geq 1$. By (3) there are $0 < c' \leq C'$ and n_0 with $c'9^n/n^3 \leq f_n \leq C'9^n/n^3$ for $n \geq n_0$. Put $c = \min\{c', \min_{1 \leq n < n_0} f_n n^3/9^n\} > 0$ and $C = \max\{C', \max_{1 \leq n < n_0} f_n n^3/9^n\}$. Then

$$c \frac{9^n}{n^3} \leq f_n \leq C \frac{9^n}{n^3} \quad (n \geq 1).$$

The series f has nonnegative coefficients and radius of convergence $1/9$, so it is analytic at the origin. Lemma 6.3 of Bóna and Burstein [9], with $K = 9$ and the integer $m = 3 > 1$, shows that f is not an algebraic power series. As $B = f - z - z^2 - z^3$ differs from f by a polynomial, B is not algebraic either. $\square$

The source does not state this consequence of its theorem. Report 136 of the same bundle (in `a217057-unique-pattern-occurrences`) draws it in the same way for one 1243; Bóna and Burstein's own non-algebraicity theorem concerns one copy of $k \cdots 21$ with k even and does not cover 1432. Whether B is D-finite is open (Question 7).

7 An exact eligible-center statistic

The marked inverse suggests a finite geometric statistic on avoiders. The following refinement is logically separate from the computer-free order theorem. Its sufficiency proof uses the exhaustive certificate specified below.

Given $q \in \text{Av}_{n+2}(1432)$ with a marked position K , require

$$1 < K < n+2, \quad D = q_{K-1} < C = q_K < B = q_{K+1}, \quad \text{pos}_q(C-1) < K-1, \quad \text{pos}_q(C+1) > K+1. \quad (18)$$

The last conditions include existence of the adjacent values. Unsplit as in Section 5. The result p has distinguished tail b, c, d at positions $j < k < l$. Require exactly one $a < d$ before j , and let its position be i . Finally require (R2)–(R4) in p . Together with sole support (R1), these are four singleton rectangles. Call such a center *eligible*.

Theorem 7.1 (Exact image with a finite certificate). *The split is a bijection between permutations of size n with one 1432 and eligible marked centers of 1432-avoiders of size $n+2$. In particular,*

$$b_n = \sum_{q \in \text{Av}_{n+2}(1432)} \#\{K : K \text{ is eligible in } q\}.$$

Proof. Necessity follows from the split and (R1)–(R4). Conditions (18) ensure that all roles are distinct. Unsplit then split again: deleting D, B and replacing the $C-1, C+1$ entries removes exactly the two new ranks adjacent to C , while deleting their neighbors removes exactly the two new positions adjacent to K . After standardization the forward construction therefore recovers the given q and its mark. In particular p has the designated occurrence (i, j, k, l) .

Suppose p has another occurrence. Retain the union of this occurrence and the designated core, at most eight points, and standardize. All four singleton-region conditions persist: their mandatory core points remain, and deletion introduces no extra points. Splitting commutes with this restriction. To see this, give X, B ancestor b , give D, Y ancestor d , and give every other split point its own old ancestor. Restriction to a set containing the core keeps precisely the output points with ancestors in that set. The two new values remain immediately adjacent to c among old values;

the two new positions remain immediately adjacent to k among old positions. Every comparison is preserved. Thus splitting the restricted source and standardizing equals restricting the split output and standardizing, and is still an avoider.

It remains to check the following finite, completely specified statement: for each $4 \leq m \leq 8$, every permutation $p \in S_m$ having at least two occurrences, and every designated 1432 core passing (R1)–(R4), has a split containing 1432. The companion enumerates these objects in two independently written implementations, one using word substitution and one using integer plot coordinates. Both give the following counts; all designated cores counted in the middle row have a nonavoiding split. The sealed data includes every one of these 2,074 designated cores, its exact split, and one explicit forbidden quadruple in that split; replay independently enumerates the complete list again.

Size m	4	5	6	7	8
Cores in multiple-occurrence sources	0	0	6	130	1938
Cores with an avoiding split	0	0	0	0	0

This finite certificate contradicts the reduced counterexample and proves sufficiency in every size. The inverse already proved gives uniqueness. $\square$

Sole support does not suffice without the other three regions. For $q = (1, 3, 6, 2, 4, 7, 5)$ with one-based center 5, the inverse is $p = (1, 5, 4, 3, 2)$, which has four occurrences. The exact eligible-center statistic provides a possible target for amplitude analysis, but no limiting density for this statistic is established here.

8 Prior results and reproducibility

The exponential growth rate 9 is inherited from avoidance and the general fixed-occurrence theory of Mansour, Rastegar and Roitershtein [8]. Their proof of Theorem 3.1 displays the stronger one-step estimate $f_m^\xi(n) \leq n \sum_{r=0}^{m-1} f_r^\xi(n-1)$, where $f_r^\xi(n)$ counts exactly r occurrences of ξ . For $m = 1$ and $\xi = 1432$, this would give $b_n \leq nA_{n-1}$ and hence the same upper order. We do not rely on that estimate here: the marked split independently proves the upper bound actually used in Theorem 1.1. No novelty is claimed for the upper order. The contribution established by the argument here is the complete stated order theorem with a matched lower scale and explicit constants, relative to the conjecture being addressed. A scoped literature and repository check is not a global priority proof.

Remark 8.1 ([write] The one-step bound of Mansour, Rastegar and Roitershtein, 5 October 2026). The display quoted above is printed in the proof of Theorem 3.1 of [8] (arXiv version 1, p. 29; the same text in the published author manuscript), where it is derived from one sentence: removing the leftmost letter of the leftmost occurrence of ξ in $\pi \in S_n$ and renaming the remaining letters gives $\pi' \in S_{n-1}$ with at most $m-1$ occurrences. That map recovers π from π' only together with the position *and the value* of the removed letter, so the printed argument bounds each fibre by n^2 , not by n , and proves

$$f_m^\xi(n) \leq n^2 \sum_{r=0}^{m-1} f_r^\xi(n-1).$$

This suffices for Theorem 3.1 there, which concerns exponential rates only. For $\xi = 1432$ and $m = 1$ it gives $b_n \leq n^2 A_{n-1} = O(9^n n^{-2})$, one power of n short of the upper order in (3).

The fibres do exceed n . For $n = 8$, let $v \in \{1, 2, 3\}$ and $p \in \{1, 2, 3\}$, and let π be obtained from $4571263 \in \text{Av}_7(1432)$ by raising every letter $\geq v$ by one and inserting v at position p ; the nine

permutations are

15682374, 25681374, 35681274, 51682374, 52681374, 53681274, 56182374, 56281374, 56381274.

A 1432 is a letter followed by a decreasing triple of larger letters. In each π the letters other than v read 5, 6, 8, x , y , 7, 4 in this order, where $\{x, y\} = \{1, 2, 3\} \setminus \{v\}$ and $x < y$. No letter other than v starts a 1432: the larger letters following 5, 6, x and y are among 6, 8, 7, among 8, 7, the letters y , 7, 4 and the letters 7, 4, respectively, and 8, 7 and 4 are followed by no larger letter; none of these lists contains a decreasing triple. The larger letters following v are, in order, some of 5, 6, then 8, then some of x , y , then 7, 4; as $x < y < 4$, their only decreasing triple is (8, 7, 4). Hence π has exactly one 1432, namely $(v, 8, 7, 4)$; its leftmost letter is v , and removing it gives back 4571263. So this avoider of size 7 has at least nine preimages in S_8 . By exhaustive enumeration the largest fibre is 2, 4, 6, 9, 12 for $n = 5, \dots, 9$.

Whether the stated inequality with the factor n holds is therefore not settled by [8]; it is Question 4. The source’s sentence that the estimate “would give” the same upper order is correct as a conditional, and its non-claim of novelty for the upper order is kept as the source’s own. But among the sources the manuscript cites, the marked split of Section 5 is the only complete proof of $b_n = O(9^n/n^3)$ that the write found. (The placement record of this batch, commit `e85586b7c`, says that the upper order was known through this paper; this remark qualifies that statement.)

The original Backelin–West–Xin journal full text was not successfully retrieved in this investigation. Attribution to that work is distinct from the directly checked proof in Krattenthaler’s Theorems 1–3, whose hypotheses and rectangle convention were verified above. The cited Burstein paper concerns a unique global decreasing triple; it does not by itself prove Lemma 5.1. Conway–Guttmann Section 5.5 repeats identifier A224182 for a different class; the identification here uses OEIS’s 2341 definition and reversal, consistent with their Section 5.3 and Nakamura’s equation.

8.1 What the companion checks

The offline archive contains this editable L^AT_EX article and PDF, two standard-library-only exact implementations, closed typed data, a read-only integrity/replay/build tool, and source provenance. It needs no network access or third-party Python package. Normal and optimized isolated Python runs must agree byte for byte; assertions removable by `-O` are not used as checks.

Through size eight the replay checks every permutation and all 5,102 one-occurrence sources, the marked inverses and injectivity, and the exact equality for each minima position-and-value skeleton. It also checks 75,419 core-preserving restriction commutations, all 110,459 interior-center candidates in avoiders of sizes six through eight, all 314 marked-minimum inflations through source size five, the explicit unmarked collision, the 106 versus 107 board obstruction, and the finite exact-image certificate. Mathematical fixture values are recomputed rather than merely trusted after a hash comparison.

The main theorem does not depend on these finite tests: its split proof, inflation argument, entropy estimate, and uses of cited avoidance theorems are ordinary mathematical proofs. Theorem 7.1 *does* use its finite exhaustive certificate, after a proved universal obstruction reduction. This distinction is essential. Neither the numerical cross-checks nor the archive seal constitute formal verification, a digital signature, or proof of an asymptotic amplitude.

The companion’s documented commands reject nonisolated launches before nonbuilt-in imports, enforce a closed inventory and strict JSON schemas, reject unsafe output targets and symlink paths, compare exact replay in normal and optimized modes, and test deliberately corrupted copies. PDF reproduction performs two fresh clean builds and compares their bytes with the frozen PDF.

Deterministic ZIP packing is checked again after fresh extraction. These are reproducibility and integrity controls, not a substitute for reviewing the mathematics.

[write] **The companion here, and the intake reruns, 5 October 2026.** “The offline archive” is the delivered package; in this report its four programs are `code/verify.py`, `code/primary.py`, `code/coordinate.py` and `code/replay.py`, its sealed data `checks/fixtures.json` is `data/checks-fixtures.json` (864,678 bytes, the 2,074 certificate records and the other fixture values), and its source provenance is `SOURCES.md`; the PDF and the manifest are not shipped. The verifier enforces the delivered layout (it reads `manifest.json`, `checks/` and `Report139.pdf`), so it does not run in the shipped directory; the report README gives two routes, from the arrival commit’s archive and from the shipped programs. At placement, on copies, `check`, `replay` and `-0 replay` passed (43–49 s under load; both outputs byte-identical to the sealed fixture) and `selftest` passed 51 tests in 238 s. At the write, on a fresh extraction, `check` and `replay` (normal and `-0`) passed again in about 5 s each, with output byte-identical to the shipped fixture, and running `replay.run()` from copies of the three shipped mathematical programs reproduced the shipped fixture’s content exactly. The PDF build and packing commands, which compare TeX environments, were not run.

9 Further questions

The principal unresolved question is whether $b_n/(nA_n)$ has a limit, and if so whether that limit is the suggested $3/80$. The exact image formula in Theorem 7.1 turns this into a question about the asymptotic mean number of eligible centers in a uniform avoider. Even a proof of convergence without evaluation would strengthen the order theorem.

The constants $1/2187$ and 81 arise from transparent injections and are not asserted to be sharp. Better control of the split image could narrow them and the inverse bracket. For $1 \oplus \delta_k$, a matching general upper bound would complement Corollary 4.2; none follows from the present eight-case proof. Finally, a computer-free sufficiency proof for the four-rectangle exact image criterion would remove the finite certificate from that additional theorem, just as the eight-case argument already does for the main upper bound.

9.1 Further questions and research

[write] **Added 5 October 2026, batch 105.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. The three paragraphs above are Questions 1, 2 (first paragraph), 3, 5 (second) and 6 (third); Questions 4 and 7 come from Remarks 8.1 and 6.3; Question 8 concerns the external inputs. The intake found no false claim in the source.

1. **Does $b_n/(nA_n)$ converge, and to $3/80$?** (Source, first paragraph; Conway and Guttman [3, Section 5.3].) Theorem 1.1 confines the ratio to $[1/2187, 81]$ in the limit; nothing in the source bears on convergence. Missing: any limit theorem; Question 2 is one route.

Observation, not a claim. From the 18 terms of A224182 and exact A_n (hook-length formula), the ratios $r_n = b_n/(nA_n)$, to seven digits, are

n	10	11	12	13	14	15	16	17	18
r_n	.0369735	.0375056	.0378384	.0380455	.0381713	.0382435	.0382801	.0382927	.0382892

They increase through $n = 17$ and decrease for the first time at $n = 18$ ($r_{18} - r_{17} = -3.5 \cdot 10^{-6}$); r_n first exceeds $3/80 = 0.0375$ at $n = 11$. Polynomials in $1/n$ of degrees 1, 2, 3, 4 through the last 2, 3, 4, 5 terms extrapolate to 0.0382, 0.0361, 0.0367, 0.0370; for degrees 2–4 the

extrapolated value increases as the last term used moves from $n = 14$ to $n = 18$. All of this is consistent with Conway and Guttmann's 0.0375 ± 0.0005 , which used 26 further approximate terms. The placement record of this batch (**e85586b7c**) and its dossier read the same data as “above $3/80$ and rising”; the turn at $n = 18$ corrects that reading. Eighteen terms cannot separate $3/80$ from nearby values, and none of these numbers is a bound.

2. **A limiting density of eligible centres** (source, first paragraph and after Theorem 7.1). By Theorem 7.1, $b_n = A_{n+2}E_{n+2}$, with E_{n+2} the mean number of eligible centres of a uniform 1432-avoider of size $n + 2$. Since $A_{n+2}/A_n \rightarrow 81$, the ratio $b_n/(nA_n)$ converges if and only if E_{n+2}/n does, and the value $3/80$ corresponds to $E_{n+2} \sim n/2160$. Missing: any law of large numbers for eligible centres.
3. **Sharper constants and a narrower inverse bracket** (source, second paragraph). The constants $1/2187$ and 81 come from the two injections and are not claimed sharp; the unrounded width $11/2$ of Corollary 6.1 is $\log(81 \cdot 2187)/\log 9$. If the inequality of Question 4 holds, the upper constant improves from 81 to $1/9$ ($C_+ = C_0/9$) and the width to $\log(2187/9)/\log 9 = 5/2$. Missing: control of the split image (the eligible-centre statistic) or a better lower injection.
4. **The one-step inequality with the factor n** (from Remark 8.1). Is $f_m^\xi(n) \leq n \sum_{r < m} f_r^\xi(n-1)$ true, in general or for $\xi = 1432$, $m = 1$, that is $b_n \leq nA_{n-1}$? The printed argument of [8] gives n^2 , and its fibres exceed n from $n = 8$. *Observation:* $b_n/(nA_{n-1})$ increases from 0.0417 at $n = 4$ to 0.2787 at $n = 18$; brute force over every m , $n \leq 7$, for the patterns 123, 132, 1234, 1243, 1324, 1432 and 2413 finds no violation (largest ratio 0.46). If Conway and Guttmann's estimate is right, $b_n/(nA_{n-1}) \rightarrow 9 \cdot 3/80 = 0.3375$, inside the bound. Missing: a proof; counting the fibres of the deletion map cannot give it.
5. **A matching upper bound for $\tau_k = 1 \oplus \delta_k$, $k \geq 4$** (source, second paragraph; Corollary 4.2). The lower bound $\liminf b_n^{(k)}/(nA_n^{(k)}) \geq k^{-2k-1}$ is proved; no upper bound of order $nA_n^{(k)}$ is. The eight-case proof of Lemma 5.1 is specific to $k = 3$, and the printed argument of [8] gives only $b_n^{(k)} \leq n^2 A_{n-1}^{(k)}$, a factor n too large. Missing: a split for a longer decreasing tail, with its case analysis.
6. **A computer-free proof of sufficiency in Theorem 7.1** (source, third paragraph). The reduction to at most eight points is proved; the remaining statement is the finite certificate of $0 + 0 + 6 + 130 + 1938 = 2,074$ designated cores in sizes 4–8. Missing: a case analysis, like Section 5's, showing that a second occurrence always survives the split.
7. **Is the generating function of A224182 D-finite?** (The delivered README: the package “does not prove ... D-finiteness”; Remark 6.3.) It is not algebraic. The integer exponent -3 does not obstruct D-finiteness, and Nakamura's functional equation (FE2341) [2] uses $n(n+1)$ catalytic variables at size n . Missing: an annihilating operator, or an argument excluding one.
8. **External inputs and priority.** The proofs use Regev's strip asymptotic [7] and Krattenthaler's Theorems 1–3 [6], which the write located but did not re-derive, and attribute the shape-Wilf ingredient to Backelin, West and Xin [5], whose journal text neither the source nor the write read. The source's literature check was scoped; priority for the matched order and the constants is not established beyond it.

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